

Siamese-CoAtNet: A Hybrid Attention Convolution for Retinal Disease Severity Assessment

S Saikiran
Department of CSE(AIML)
PES University
Bengaluru, India
PES1202203614@pesu.pes.edu

Vishruth H V
Department of CSE(AIML)
PES University
Bengaluru, India
PES120220368@pesu.pes.edu

Uttam G Raj
Department of CSE(AIML)
PES University
Bengaluru, India
PES1202203595@pesu.pes.edu

Preethi S J
Department of CSE(AIML)
PES University
Bengaluru, India
preethisj@pes.edu

Abstract—A significant cause of preventable loss of vision in the world is Diabetes related retinal disorders. Therefore, it is necessary to identify retinal changes as soon as possible and accurately define the severity of the disease to be able to manage it effectively. In this work, we propose Siamese-CoAtNet, a deep learning model that leverages both the advantages of a Siamese network and CoAtNet backbone to draw out the fine-grained local structures and global large-scale patterns in fundus retinal images. We use APTOS 2019 and IDRiD datasets for training and evaluating our Siamese-CoAtNet framework. To enhance the diagnostic reliability, the network integrates image augmentations and CLAHE preprocessing in order to preserve faint yet clinically significant features and still preserve the overall integrity of the fundus image. Moreover, Grad-CAM visualisations highlight the most influential retinal regions on the prediction of the model and are hence valuable for interpretability. Collectively, these features offer a framework that is interpretable, stable, and generalisable—capable of aiding ophthalmologists with the early detection and grading the severity of DR.

Keywords — *Diabetic Retinopathy, Fundus Imaging, Siamese, CoAtNet, Grad-CAM, Medical Image Classification.*

I. INTRODUCTION

Diabetic Retinopathy (DR) is a progressive retinal disease and a major cause of global vision impairment. Timely identification and grading of DR severity are crucial for effective treatment to prevent irreversible vision loss. Traditionally, diagnosis relies on careful inspection of retinal fundus images by an expert ophthalmologist. However, this process can be time-consuming and dependent on the expertise of an ophthalmologist. To automate DR screening, several deep-learning-based models have been explored in the literature; however, existing models often suffer from limited generalizability across datasets, lack explainability, and either use small-resolution inputs or fail to provide clinical interpretability. The proposed work utilizes a Siamese network with CoAtNet-0 [11]-Base architecture, optimized with triplet loss, to perform diabetic retinopathy (DR) diagnosis on fundus images. The CoAtNet combines convolution and self-attention mechanisms to learn the representations of local lesions (e.g., microaneurysms, hemorrhages) and

features of the global retina. The Siamese network [20] with CoAtNet backbone is optimized by using a triplet-based objective which encourages similar instances to cluster closer together in the feature space and enforces sufficient distance between dissimilar cases. The proposed Siamese-CoAtNet model is trained and evaluated using the APTOS 2019 dataset [17], with cross-dataset validation conducted on the IDRiD dataset [18] to assess its generalizability. This framework leverages CoAtNet pre-trained weights (trained with Imagenet) and is fine-tuned for the DR diagnosis domain, achieving strong performance in diabetic retinopathy (DR) severity classification that aligns with expert annotations.

II. LITERATURE REVIEW

Researchers prefer automated DR screening systems due to their ability to accurately identify the severity levels of DR by leveraging expert-labeled data. In this era of deep learning, many studies are conducted using deep learning models.

Gulshan et al. [1] devised and confirmed a CNN-less model which performs automated analysis of fundus images, achieving high sensitivity and specificity. However, they primarily provided a clinical validity and evaluation benchmark extrapolation based on binary grading in retinal images without external validation on a large, heterogeneous population. Pratt et al. [4] developed classification with a CNN using transfer learning, which can provide better classification ability and lowered training time. Regardless, both studies again outlined the issues with generalizability and clinical validity compared to the external datasets in handling real life cases.

Alyoubi et al. [2] highlighted works using deep learning for DR detection and grading as presented, the improvements to CNN architecture, data augmentation, and preprocessing and also challenges such as, heterogeneity of dataset, class imbalance and variability with benchmarks. Bi et al. [8] accomplished a systematic review and meta-analysis of the 47 studies, and confirmed models typically achieved high sensitivity and specificity, and that

there is still standardization, interpretability and external validation that needed to be addressed. Both reviews confirmed notable progress was made while still recognizing the challenges with deep learning as it relates to DR research.

Mutawa [3] developed a hybrid network employing multi-scale image interpretation for dependable real-time interpretation but suffered from high computational requirements, lack of visibility, and no peer-reviewed clinical trials. Govindharaj et al. [9] designed a capsule network framework that demonstrated a better performance because it was able to exploit spatial hierarchies within the retinal images that helped detect important discriminative features like microaneurysms. Each of these approaches sought to examine alternatives to CNNs, however, the computational demands would still not be able to be utilised in a low-resource context.

Zhou [5] provided a generalizable model of ocular disease diagnosis to permit multitasking outside diabetic retinopathy (DR). detection that was time consuming, required a lot of computation, and remained difficult to interpret, but acknowledged existing limits of large and diverse dataset properties. Ai et al. [10] shifted the focus to the clinical robustness of NLP systems with an ensemble framework (DR-IIXRN) of multiple CNNs and transformers (InceptionV3 and NASNetLarge) with attention that achieved better performance than single network baselines on hospital datasets.

Uppina et al. [6] studied UNET and stacked UNET on the APTOS dataset for multi-class DR grading. They found that stacked UNET usually outperformed UNET by using features from the previous UNET. However, it faced problems with class imbalance and generalization. Sajid et al. [7] introduced DR-NASNet, a convolutional neural network that used neural architecture search with dense blocks and an SVM classifier. This network achieved good results for five-class DR grading on APTOS while focusing on computational efficiency. The examples of segmentation and neural architecture search showed that optimized networks generally performed better than standard convolutional neural networks for five-class DR classification.

Recent work has also investigated Siamese architectures for diabetic retinopathy detection. Santos et al. [21] created a Siamese CNN that learned pairwise similarity between retinal photographs. The similarity feedback allowed the model to learn small differences that are specific to severity grading between images of patients diagnosed with diabetic retinopathy across many datasets, including APTOS, IDRiD, and Messidor-1. For Tariq et al. [22], to implement a few-shot Siamese framework while using a pre-trained backbone, including VGG16, ResNet50, and DenseNet121, and generalized well with so few training examples. The method was validated on datasets like APTOS and FGADR. Together these two studies provided evidence on the effectiveness of similarity learning and highlighted how Siamese networks are complementary to traditional

classifiers in utilizing the relational properties between retinal image pairs.

III. PROPOSED METHODOLOGY

This paper proposes an explainable hybrid attention convolution framework to grade severity levels of diabetic eye condition using fundus images. A hybrid convolution-attention network, CoAtNet, is used as the backbone of a Siamese architecture, trained with triplet loss on the APTOS 2019 dataset [17] of fundus images that are labeled over five severity categories (levels) of diabetes-induced ocular abnormality. Framework's performance is assessed using both in-dataset evaluation on APTOS and cross-dataset evaluation on the IDRiD dataset [18] using both pairwise retrieval accuracy and hold-out classification accuracy to evaluate performance and generalization. The overall pipeline consists of dataset loading and preprocessing, data augmentation to mitigate class imbalance, triplet training using tuned hyperparameters, and model testing and evaluation. For qualitative evaluation, t-SNE visualisation is utilized to evaluate the separability of DR categories (levels) across datasets and the learned embedding space.

A. Dataset

APTOS 2019 dataset was employed for training and evaluating the presented Siamese-CoAtNet model to identify diabetic retinopathy. APTOS includes 3,662 color fundus images that are annotated with five severity levels. The images are captured at a variety of resolutions (usually about 3216×2136 pixels), and were taken in various real-world conditions, and have natural variability that has proven useful for learning rich features. To evaluate the model's generalization, cross-dataset validation is performed using the IDRiD dataset, which has high-quality (4288×2848 pixels) retinal fundus images taken in controlled clinical settings. Like the APTOS dataset, the IDRiD dataset has all images annotated by an experienced clinician (ophthalmologist) and scores its severity with the same five-class DR severity scale. Figures 1(a) and 1(b) show representative images from each severity level, and Table I presents the class-wise distribution across APTOS and IDRiD.

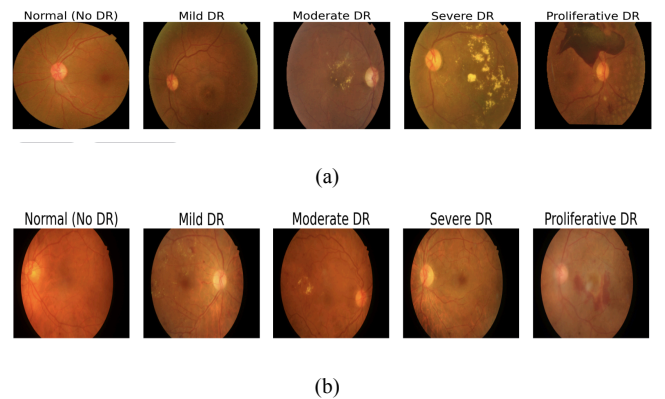


Fig. 1. Samples of APTOS 2019 and IDRiD fundus retinal images depicting various severity levels of diabetic retinopathy, ranging from Normal (No DR) to Proliferative DR.

TABLE I. DESCRIPTION OF DATASETS

Class	#images in APTOS	#images in IDRiD
NORMAL	1805	129
MILD	370	22
MODERATE	999	156
SEVERE	193	84
PROLIFERATIVE	295	64

B. Data Preprocessing and Augmentation

CLAHE [12] is one of the preprocessing methods that aims at enhancing the local contrast in the retinal images. It divides the image into a few contextual areas, computes the intensity distribution of each and then equalizes the pixels individually to boost visual sharpness without amplifying noise too much. This technique successfully enhances the visibility of important clinical characteristics such as microaneurysms and hemorrhages while reducing noise levels. As depicted in Fig. 2, CLAHE significantly enhances visibility in darker areas, which is beneficial for improving classification performance at the earliest stages of diabetic retinopathy.

To address data imbalance and enhance the robustness of the model, a number of augmentation strategies including horizontal mirroring, adjustment of brightness and minor rotations were implemented on the training samples. Also, Gaussian noise and ISO noise, as well as blurring, were introduced. The goal is to incorporate realistic variability without compromising important anatomical structures. These augmentations mimic natural variations of device settings and orientations that would be encountered in realistic conditions. Thereby, a more diverse training dataset is generated, making the model less prone to overfitting.

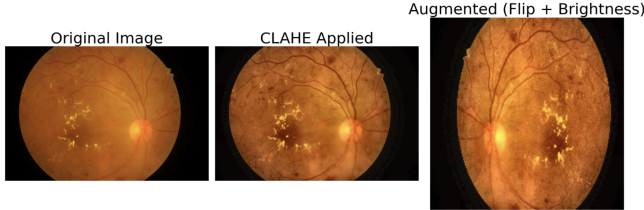


Fig. 2. Data Preprocessing and Augmentation of a sample from the Moderate DR class

C. Model Architecture

The framework developed in this study employs CoAtNet [11] as the base architecture, which was adapted into a Siamese configuration. CoAtNet combines convolutional layers with self-attention layers. CLAHE preprocessing is applied to the RGB images before augmentation, after which they are rescaled to dimensions of 224×224 and are normalized with respect to the ImageNet distribution statistics and converted into PyTorch tensors. The architecture contains five stages as shown in Fig. 3. Stages S0, S1, and S2 extract low- and mid-level features (convolutional blocks), while transformer stages S3 and S4 capture long-range dependencies, patterns of lesions, and abnormalities at a global scale. The dense layers combined with the global pooling component and a softmax generate a probability distribution over the five grades of

diabetic retinopathy.

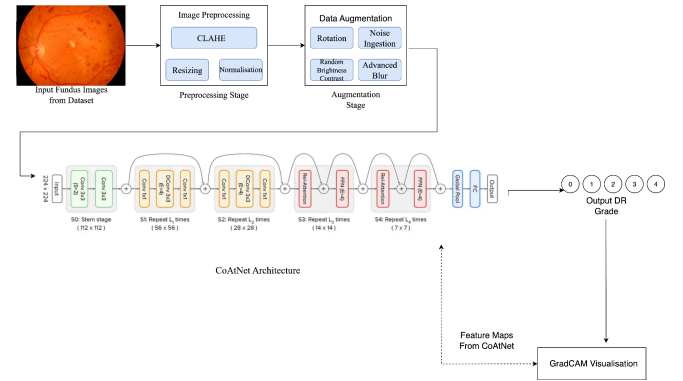


Fig. 3. Flowchart of the proposed methodology for DR grading using CoAtNet.

Our model is trained to recognize subtle but important retinal image differences that suggest diabetic retinopathy severity. For this, we utilize a Siamese network architecture derived from CoAtNet, a strong image understanding model. We use the CoAtNet-0 variant specifically as a shared encoder, wherein the same model is employed for all images, thus producing compact, 128-dimensional feature representations that are L2-normalized for consistency. The encoder has already been learned on the APTOS dataset, including a large retinal image collection, allowing the learning of informative eye disease-related patterns. The pretrained weights are used as is in our Siamese configuration, and the encoder parameters are left in a frozen state so that the model maintains its learned features up to this point. We add a small learnable projection layer as well, which projects the encoder output into a more appropriate form for our particular task. At training time, the model is presented with three images simultaneously: a reference image (anchor), a visually similar image (positive), and a dissimilar image (negative). By contrasting these images, the model learns to move similar images closer together and dissimilar images apart in feature space with a method known as triplet loss. The pipeline for training the Siamese network with CoAtNet base is illustrated in Fig. 4.

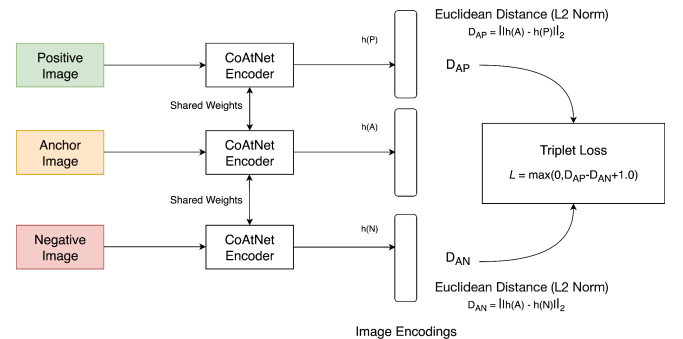


Fig. 4. Flowchart of the Siamese-CoAtNet based pipeline for DR grading

D. Model Training and Hyperparameter Settings

Experiments are carried out on an NVIDIA Tesla P100 GPU while using Python and PyTorch for model

development; image processing is conducted using both OpenCV and Albumentations. For baseline CoAtNet training, the IDRiD dataset is split 80/20 for training and evaluation; the APTOS dataset is assessed using five-partition cross-validation to minimize bias and enhance generalization. Training runs for 100 epochs, utilising Adam optimizer, with a starting learning rate of 0.001, decayed to 1e-6, and using early stopping if improvement is not seen within 20 epochs. During training, CoAtNet is compared against ResNet50 [13], InceptionV3 [14], and MobileNet [15]. Accuracy and quadratic kappa scores are consistently higher than those of the other architectures, so CoAtNet is chosen as the base for Siamese-CoAtNet. GradCAM [16] is applied to the final convolutional-attention blocks, producing heatmaps that highlight clinically relevant regions such as microaneurysms, hemorrhages, and exudates, providing interpretability for ophthalmologists.

The Siamese-CoAtNet model trains with a triplet-based learning framework that separates diabetic retinopathy (DR) severity levels within embedding space. Each training sample includes a reference sample, a semantically similar class sample, and a dissimilar class sample. In the pipeline, images are resized, normalized, sampled to produce 624 embeddings, and converted to tensors. Training uses Triplet Margin Loss (margin = 1.0) and Euclidean distance for the similarity metric. The Adam optimizer with a learning rate of 1e-4, combined with a ReduceLROnPlateau scheduler (patience = 4, factor = 0.5), guides optimization. The model is trained for 40 epochs with a batch size of 32 samples each. Pairwise accuracy—where the anchor embedding is closer to the positive than the negative—is added as an additional metric, and the model with the lowest average triplet loss is saved and used for evaluation.

IV. EXPERIMENTAL RESULTS

A. Performance Metrics

Model performance is assessed by accuracy, weighted F1, precision, and recall. Cohen's Kappa [19], which considers the predicted and actual labels, indicates agreement beyond chance and is defined as:

$$Kappa\ Score = \frac{OA - EA}{1 - EA} \quad (1)$$

where OA is the share of predictions that match the ground truth, and EA is the expected agreement from chance. To examine if the model generalizes the similarities it has learned from the training images into unseen images, the triplet images were held out from the training set. The Siamese-CoAtNet model employs pairwise accuracy to have some measure of whether or not the anchor embedding is closer to the positive samples than to the negative samples. Holdout accuracy looks to measure performance in triplet images that were not observed during training time. Both of these accuracy are used in assessing model performance.

B. Performance Analysis

The CoAtNet architecture is tested using the IDRiD dataset which contains high-resolution fundus images that have

been labeled by expert clinicians at all 5 stages of diabetic retinopathy. The IDRiD dataset is ideal for a real-world comparison because it captures actual patient classifications with imbalanced class distribution and large intra-class variance. First, a baseline version of CoAtNet-0 was trained on the IDRiD dataset showing a validation Kappa score of 0.9024 with the corresponding validation accuracy of 71.43%, which indicated strong agreement with clinical labels. Next, 5-fold cross-validation was conducted on the APTOS 2019 dataset to highlight the ability of the model to generalize. Table II shows results from the literature which have used the APTOS dataset, and Table III shows results from the 5-fold evaluation of the CoAtNet-based architecture.

TABLE II. EXISTING RESEARCH ON APTOS 2019 DATASET

Model	Kappa Score	Accuracy
ResNet50[13]	0.78	0.74
InceptionV3[14]	0.79	0.78
MobileNet[15]	0.77	0.84
Proposed Method (CoAtNet)	0.91	0.85

TABLE III. 5-FOLD CROSS-DATASET VALIDATION ON APTOS

CoAtNet Model	Kappa Score	Accuracy	Train Accuracy	Train Loss
1st Fold	0.9113	0.8417	0.8822	0.0170
2nd Fold	0.8983	0.8281	0.9891	0.0015
3rd Fold	0.9053	0.8538	0.9812	0.0023
4th Fold	0.9061	0.8402	0.9805	0.0024
5th Fold	0.8942	0.8374	0.9659	0.0056

The averaged, multiclass ROC curves over the five folds for DR classification are shown in Figure 5. Each level has a quite high AUC score (ranging from 0.91 to 1.00), hence the model is highly confident in differentiating levels of this disorder.

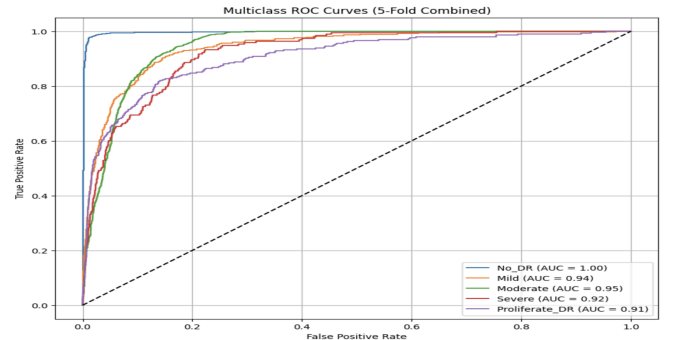


Fig. 5. Multiclass ROC Curves for DR Severity Classification

CoAtNet-Siamese framework starts with the first

training of a CoAtNet-0 model as a standalone classifier on the APTOS 2019 dataset. This dataset includes high-definition retinal fundus imagery labeled across five severity levels of diabetic retinopathy. Around 80% of the samples are used to train the model and the rest for validation through a stratified sampling strategy for preserving class distribution in training and validation splits. It is run for 40 epochs using AdamW optimisation technique, a CosineAnnealingWarmRestarts learning rate scheduler and Focal Loss employing class-balanced weight factors to compensate for class imbalance, using a learning rate of $3e-4$ along with weight decay of $1e-4$, and early stopping was adopted based on the validation Kappa score. The best checkpoint (.pth) of the model identified by its peak validation Kappa and accuracy is then saved for downstream applications.

In a Siamese network, the pretrained CoAtNet-0 model operates as a shared encoder that extracts embeddings from triplets of anchor, positive and negative images. The frozen encoder retained the previously learned representations during the training of CoAtNet. A small learnable projection head mapped features into a normalized embedding space of dimension 128. During training, a trainable batch size of 32, the Adam optimizer, ReduceLROnPlateau scheduler and triplet loss was used for a total of 40 epochs to minimize distance between positive pairs of images and negative pairs of images. Performance was evaluated using pairwise accuracy, holdout accuracy and the traditional metrics of precision, recall and F1-score. The Siamese-CoAtNet achieved 93.94% pairwise accuracy, 91.95% holdout accuracy, 0.1885 triplet loss, 0.9581 F1-score, and 0.9195 recall on the APTOS 2019 dataset, indicative of effective learning of the semantic similarity and class-wise separation. To evaluate whether the pretrained CoAtNet checkpoint utilized in the original model could generalize, the trained model was evaluated on the IDRiD dataset directly from checkpoint without fine-tuning and achieved 79.95% pairwise accuracy, 72.53% holdout accuracy and triplet loss of 0.2490, indicating the model was able to transfer learned representations from the APTOS data to the IDRiD dataset which had domain variation. Table IV shows the results obtained while utilizing the Siamese-CoAtNet framework.

TABLE IV. SIAMESE-COATNET RESULTS ON APTOS & IDRiD

Siamese-CoAtNet Framework Results		
	APTOS Dataset	IDRiD Dataset (Cross-Validated)
Pairwise Accuracy	0.9394	0.7995
Triplet Loss	0.1885	0.2490
Hold-Out Accuracy	0.9195	0.7253

TABLE V. COMPARISON WITH EARLIER WORKS UTILISING THE SIAMESE NETWORK ARCHITECTURE.

	Acc.	Recall	F1 Score	Training time (min:sec)	Testing Time (min:sec)
Tariq et al. [22]	0.81	-	-	-	-
Santos et al. [21]	0.6723	0.6400	-	06:31	00:02
Ours	0.91	0.9195	0.958	01:14	00:04

Table V shows a comparison table of past studies that utilized the Siamese network architecture to identify diabetic retinopathy, with the main performance indicators (i.e., accuracy, recall, F1 score, and training and test times) presented in the current literature.

A t-SNE projection shown in Fig 6. of the Siamese-CoAtNet embeddings on the APTOS 2019 dataset shows good separation between diabetic retinopathy (DR) severity classes while exhibiting strong grouping of the classes. Class 0 forms a tight cluster, Classes 1 and 2 are well separated, showing good discriminative performance, but considerable overlap occurs between Classes 3 and 4 due to clinical and perceptual similarities between severe and proliferative DR that even an ophthalmologist struggles to differentiate. Despite this overlap, the separability in the embedding space indicates that the model learns semantically meaningful and discriminative features as expected. To understand the model's areas of focus, Grad-CAM [16] is employed, producing visual explanations that highlight the retinal regions most important for classification. Grad-CAM visualizations in Fig 7. show that the model attends to clinically significant features, covering small aneurysmal spots, hemorrhagic areas, and lipid exudate formations, suggesting that the model learns to focus on pathology-related features at a local level.

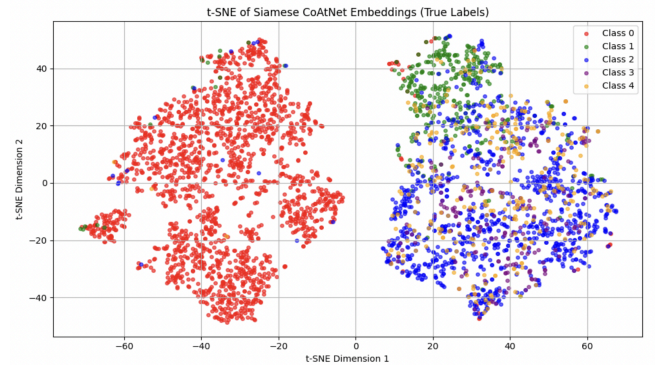


Fig. 6. t-SNE visualization of 128-dimensional embeddings generated by the Siamese CoAtNet model, with points coloured by true DR class labels.

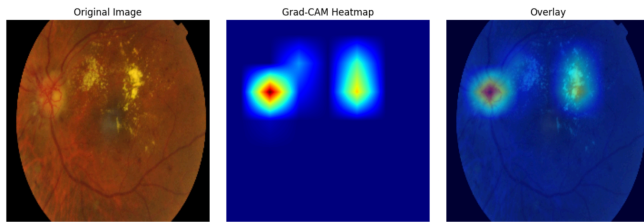


Fig. 7. Grad-CAM visualization showing the original fundus image, model attention heatmap, and overlay.

V. CONCLUSION

A deep learning framework based on Siamese-CoAtNet is presented for classification of diabetic retinopathy (DR) severity. The model is trained on the APTOS (2019) dataset, and cross-dataset validation has been applied to IDRiD to test the generalization ability and the model robustness. Comparisons to other backbones, such as ResNet50, InceptionV3, and MobileNet, indicate that CoAtNet is a good backbone for DR classification. The findings suggest that Siamese-CoAtNet is an interpretable and generalisable technique, and it can be leveraged to aid ophthalmologists and to make diagnosis easy with early detection and assessment of DR severity. Future work may explore reduced-complexity models with comparable performance and examine multimodal sources such as patient history or counseling data to further improve performance. For instance, we could implement trained LLMs for automatically generating screening reports within the system.

REFERENCES

- [1] Gulshan V. et al., Deep learning approach developed and validated for automated diabetic-retinopathy detection in retinal images, 2016.
- [2] W. L. Alyoubi, W. M. Shalash, and M. F. Abulkhair, "Diabetic retinopathy detection through deep learning techniques: A review," *Informatics in Medicine Unlocked*, vol. 20, p. 100377, 2020, doi: 10.1016/j.imu.2020.100377.
- [3] A. M. Mutawa et al., "Advanced Detection of Diabetic Retinopathy: Employing Hybrid Deep Neural Networks and Multi-Scale Image Analysis Techniques," *Preprints*, Jan. 2024, doi: 10.20944/preprints202401.2082.v1.
- [4] H. Pratt, F. Coenen, D. M. Broadbent, S. P. Harding, and Y. Zheng, "Convolutional neural networks for diabetic retinopathy," *Procedia Computer Science*, vol. 90, pp. 200–205, 2016, doi: 10.1016/j.procs.2016.07.014.
- [5] Y. Zhou et al., "A foundation model for generalizable disease detection from retinal images," *Nature*, vol. 620, pp. 774–781, Aug. 2023, doi: 10.1038/s41586-023-06555-x.
- [6] Uppina A., Navaneetha Krishnan S., Teja T. K. S., Iyer N. N., and Dhanith P. R. J., Comparative study on CNN-based UNET and stacked UNET models for diabetic-retinopathy detection, 2025.
- [7] M. Z. Sajid, M. F. Hamid, A. Youssef, J. Yasmin, G. Perumal, I. Qureshi, S. M. Naqi, and Q. Abbas, "DR-NASNet: Automated system to detect and classify diabetic retinopathy severity using improved pretrained NASNet model," *Diagnostics*, vol. 13, no. 16, p. 2645, 2023. doi: 10.3390/diagnostics13162645.
- [8] Z. Bi, J. Li, Q. Liu, and Z. Fang, "Deep learning-based optical coherence tomography and retinal images for detection of diabetic retinopathy: A systematic and meta analysis," *Frontiers in Endocrinology*, vol. 16, Art. no. 1485311, Mar. 18, 2025. doi: 10.3389/fendo.2025.1485311.
- [9] I. Govindharaj, R. Rampriya, G. Michael, S. Yazhinian, K. D. Kumar, and R. Anandh, "Capsule network-based deep learning for early and accurate diabetic retinopathy detection," *International Ophthalmology*, vol. 45, no. 1, p. 78, 2025, doi: 10.1007/s10792-024-03391-4.
- [10] Z. Ai, X. Huang, Y. Fan, J. Feng, F. Zeng, and Y. Lu, "DR-IIXRN: Detection algorithm of diabetic retinopathy based on deep ensemble learning and attention mechanism," *Frontiers in Neuroinformatics*, vol. 15, Art. no. 778552, 2021, doi: 10.3389/fninf.2021.778552.
- [11] Z. Dai, H. Liu, Q. V. Le, and M. Tan, "CoAtNet: Marrying convolution and attention for all data sizes," *arXiv preprint arXiv:2106.04803*, Jun. 2021. [Online]. Available: <https://arxiv.org/abs/2106.04803>
- [12] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego: Academic Press Professional, 1994, pp. 474–485.
- [13] He K., Zhang X., Ren S., and Sun J., Deep residual learning for image recognition, CVPR, 2016.
- [14] Szegedy C., Liu W., Jia Y., Sermanet P., Reed S., Anguelov D., Erhan D., Vanhoucke V., and Rabinovich A., Going deeper with convolutions, CVPR, 2015.
- [15] Howard A. G., Zhu M., Chen B., Kalenichenko D., Wang W., Weyand T., Andreetto M., and Adam H., MobileNets: Efficient CNNs for mobile vision, 2017.
- [16] Selvaraju R. R., Cogswell M., Das A., Vedantam R., Parikh D., and Batra D., Grad-CAM: Visual explanations from deep networks via gradient-based localization, ICCV, 2017.
- [17] APTOS Consortium. Image dataset used for the 2019 diabetic-retinopathy grading challenge
- [18] Porwal P., Pachade S., Kamble R., Kokare M., and Deshmukh G., IDRiD retinal image dataset for diabetic-retinopathy research, 2018.
- [19] Cohen J., A study introducing a coefficient to measure agreement on categorical data, 1960.
- [20] Bromley J., Signature verification using a Siamese neural network, 1993.
- [21] Santos M. S., Valadao C. T., Resende C. Z., and Cavalieri D. C., Predicting diabetic-retinopathy stages using a Siamese CNN, 2024.
- [22] Tariq M., Palade V., and Ma Y., Diabetic-retinopathy classification using a Siamese neural network for small datasets, 2024.

